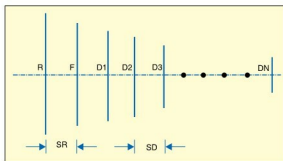


Q. Why are TV antennas (Yagi) made of tubular aluminium? Can we use any other material? How is it designed?

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A. The Yagi antenna was invented in Japan. The work was originally done by Shintaro Uda, but it was presented for the first time by Yagi. The Yagi-Uda antenna, or Yagi, is simple to construct and has a high gain, which led to its widespread use.

Elements of Yagi antenna are shown in the picture. The antenna consists of a single 'feed' or 'driven' element (F), typically a dipole or a folded dipole antenna. This is the only member of the structure that is actually



Elements of Yagi antenna

excited (a source voltage or current applied).

The element to the left of the feed element is the reflector (R). There is typically only one reflector; adding more reflectors improves performance very slightly. This element is important in determining the front-to-back ratio of the antenna.

The rest of the elements (those to the right of the feed ant-

enna are known as director elements. There can be any number of directors (N), which can typically be anywhere from 1 to 20. These elements (D1 to DN) are parasitic—they reflect or help to transmit the energy in a particular direction.

The feed antenna is always the second from the end. This feed antenna is often altered in size to make it resonate in the presence of the parasitic elements (typically, 0.45-0.48 wavelength-long for a dipole antenna).

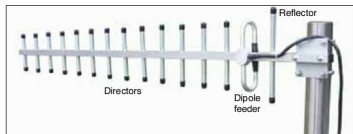
The length of the reflector is represented by R and the distance between the feed and the reflector by SR. The reflector element is typically slightly longer than the feed element. Each element is of length D_i and separated from the adjacent director by a length SD_i .

The length of the directors is typically less than the resonant length, which encourages wave propagation in the direction of the directors. The elements are generally made of aluminum tubing. The diameter of the elements is given by:

$$d=0.0085\lambda$$

Aluminum is preferred to copper and steel as it is cheaper and lighter. Use of tubular elements result in reducing the weight of the antenna. The design parameters for Yagi antenna are given in the table.

Yagi antenna has a high gain (>10 dB). It operates in high-frequency (HF) to ultra high-frequency (UHF) bands (about 3MHz to 3GHz). Bandwidth is typically small—of the order of a few per cent of the centre frequency. It has end-fire radiation pattern, medium directivity and linear polarisation.



Yagi antenna

Optimal Lengths for Yagi Elements for Distinct Boom Lengths

	Boom Length of Yagi-Uda Array (in λ)					
	0.4	0.8	1.2	2.2	3.2	4.2
R	0.482	0.482	0.482	0.482	0.482	0.475
D1	0.442	0.428	0.428	0.432	0.428	0.424
D2		0.424	0.420	0.415	0.420	0.424
D3		0.428	0.420	0.407	0.407	0.420
D4			0.428	0.398	0.398	0.407
D5				0.390	0.394	0.403
D6				0.390	0.390	0.398
D7				0.390	0.386	0.394
D8				0.390	0.386	0.390
D9				0.398	0.386	0.390
D10				0.407	0.386	0.390
D11					0.386	0.390
D12					0.386	0.390
D13					0.386	0.390
D14					0.386	
D15					0.386	
Spacing between director (SD/ λ)	0.20	0.20	0.25	0.20	0.20	0.308
Gain (dB)	9.25	11-35	12-35	14.40	15-55	16-35

Note: $D=0.0085\lambda$ (dia of the tube); $SR=0.2\lambda$

Answers compiled by Col (Retd) N.C. Pande.

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